

00_Fine_Tune_Model

June 23, 2025

1 Fine Tune YOLOv8 Model

1.1 Overview

This notebook fine-tunes a YOLOv8 model for detecting football players and their positions on the field. The model is trained to identify specific football positions and roles during pre-snap formations.

1.2 What We're Doing

1. **Dataset Preparation:** Load the “presnaps-large-model” dataset from Roboflow containing annotated football images
 - Dataset: [presnaps-large-model on Roboflow Universe](#)
2. **Model Initialization:** Start with a pre-trained YOLOv8 medium model (`yolov8m.pt`)
3. **Fine-tuning:** Train the model for 50 epochs on football-specific data
4. **Object Detection Classes:**
 - defense: Defensive players
 - oline: Offensive line players
 - qb: Quarterback
 - ref: Referee
 - running_back: Running back
 - tight_end: Tight end
 - wide_receiver: Wide receivers
5. **Model Deployment:** Push the trained model to Hugging Face Hub for easy access and sharing
 - Model: [gridiron-ai-football-detector on Hugging Face](#)

1.3 Output

The fine-tuned model will be saved and made available for football formation analysis and player position detection in game footage.

Note: This notebook was trained using a Google Colab notebook with a GPU (Tesla T4)

```
[1]: !pip install ultralytics
import ultralytics
ultralytics.checks()
```

Ultralytics 8.3.155 Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (NVIDIA
A100-SXM4-40GB, 40507MiB)
Setup complete (12 CPUs, 83.5 GB RAM, 42.0/112.6 GB disk)

```
[2]: !pip install huggingface_hub  
import huggingface_hub  
huggingface_hub.login()
```

```
Requirement already satisfied: huggingface_hub in  
/usr/local/lib/python3.11/dist-packages (0.33.0)  
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-  
packages (from huggingface_hub) (3.18.0)  
Requirement already satisfied: fsspec>=2023.5.0 in  
/usr/local/lib/python3.11/dist-packages (from huggingface_hub) (2025.3.2)  
Requirement already satisfied: packaging>=20.9 in  
/usr/local/lib/python3.11/dist-packages (from huggingface_hub) (24.2)  
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-  
packages (from huggingface_hub) (6.0.2)  
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-  
packages (from huggingface_hub) (2.32.3)  
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.11/dist-  
packages (from huggingface_hub) (4.67.1)  
Requirement already satisfied: typing-extensions>=3.7.4.3 in  
/usr/local/lib/python3.11/dist-packages (from huggingface_hub) (4.14.0)  
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in  
/usr/local/lib/python3.11/dist-packages (from huggingface_hub) (1.1.3)  
Requirement already satisfied: charset-normalizer<4,>=2 in  
/usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub) (3.4.2)  
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-  
packages (from requests->huggingface_hub) (3.7)  
Requirement already satisfied: urllib3<3,>=1.21.1 in  
/usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub) (2.4.0)  
Requirement already satisfied: certifi>=2017.4.17 in  
/usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub)  
(2025.4.26)
```

```
VBox(children=(HTML(value='<center> <img\nsrc=https://huggingface.co/front/  
assets/huggingface_logo-noborder.sv...
```

```
[3]: !pip install roboflow
```

```
Requirement already satisfied: roboflow in /usr/local/lib/python3.11/dist-  
packages (1.1.66)  
Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-  
packages (from roboflow) (2025.4.26)  
Requirement already satisfied: idna==3.7 in /usr/local/lib/python3.11/dist-  
packages (from roboflow) (3.7)  
Requirement already satisfied: cycler in /usr/local/lib/python3.11/dist-packages  
(from roboflow) (0.12.1)
```

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from roboflow) (1.4.8)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (from roboflow) (3.10.0)

Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.11/dist-packages (from roboflow) (2.0.2)

Requirement already satisfied: opencv-python-headless==4.10.0.84 in /usr/local/lib/python3.11/dist-packages (from roboflow) (4.10.0.84)

Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.11/dist-packages (from roboflow) (11.2.1)

Requirement already satisfied: pillow-heif>=0.18.0 in /usr/local/lib/python3.11/dist-packages (from roboflow) (0.22.0)

Requirement already satisfied: python-dateutil in /usr/local/lib/python3.11/dist-packages (from roboflow) (2.9.0.post0)

Requirement already satisfied: python-dotenv in /usr/local/lib/python3.11/dist-packages (from roboflow) (1.1.0)

Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from roboflow) (2.32.3)

Requirement already satisfied: six in /usr/local/lib/python3.11/dist-packages (from roboflow) (1.17.0)

Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.11/dist-packages (from roboflow) (2.4.0)

Requirement already satisfied: tqdm>=4.41.0 in /usr/local/lib/python3.11/dist-packages (from roboflow) (4.67.1)

Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.11/dist-packages (from roboflow) (6.0.2)

Requirement already satisfied: requests-toolbelt in /usr/local/lib/python3.11/dist-packages (from roboflow) (1.0.0)

Requirement already satisfied: filetype in /usr/local/lib/python3.11/dist-packages (from roboflow) (1.2.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib->roboflow) (1.3.2)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib->roboflow) (4.58.2)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib->roboflow) (24.2)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib->roboflow) (3.2.3)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->roboflow) (3.4.2)

```
[5]: from roboflow import Roboflow
      rf = Roboflow(api_key=ROBOFLOW_API_KEY)
      project = rf.workspace("boilermake-2025").project("presnaps-large-model")
      version = project.version(1)
      dataset = version.download("yolov8")
```

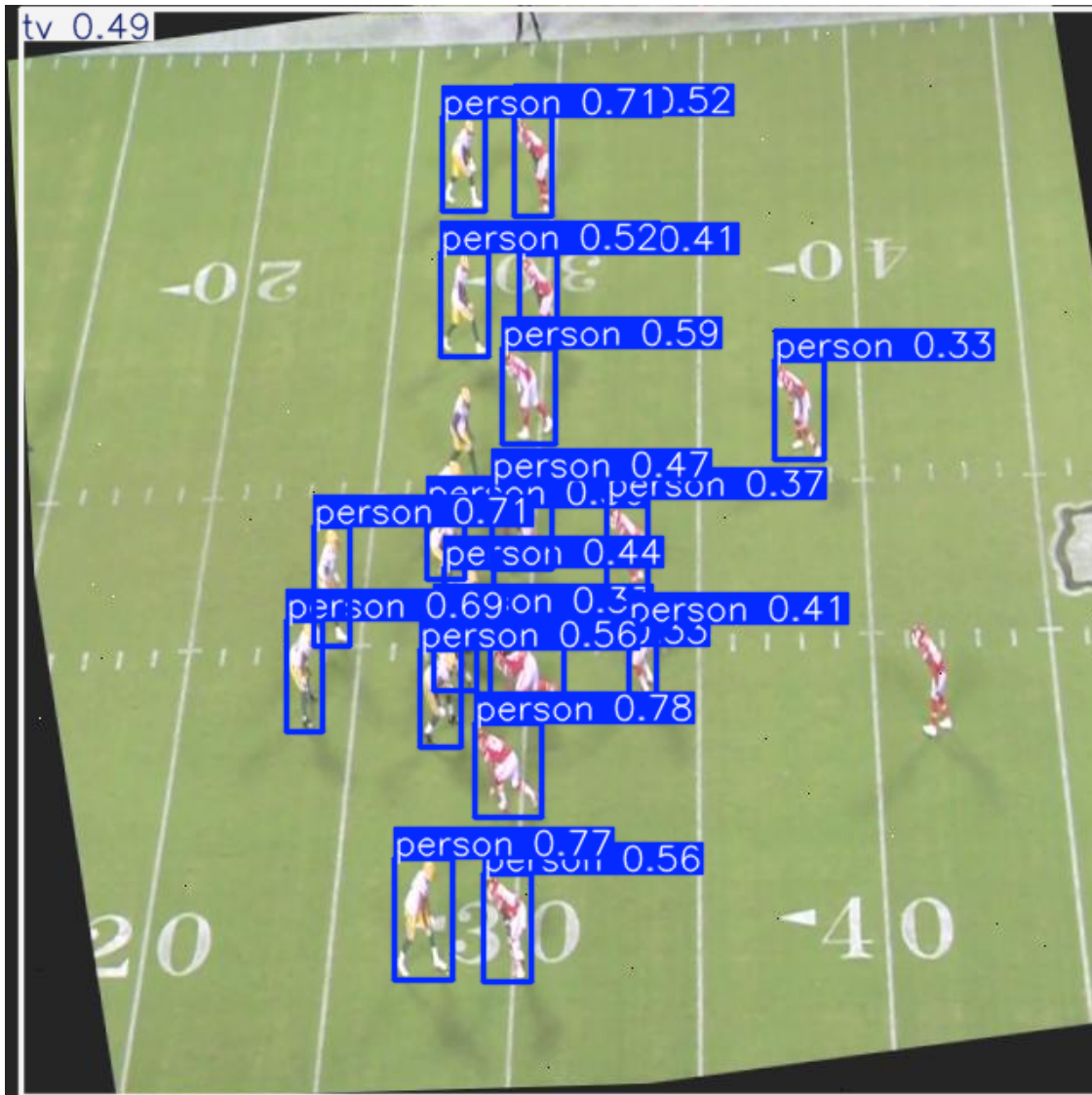
loading Roboflow workspace...

loading Roboflow project...

```
[6]: # finetune YOLOv8
      # Load pre-trained model
      from ultralytics import YOLO
      model = YOLO('yolov8m.pt')
```

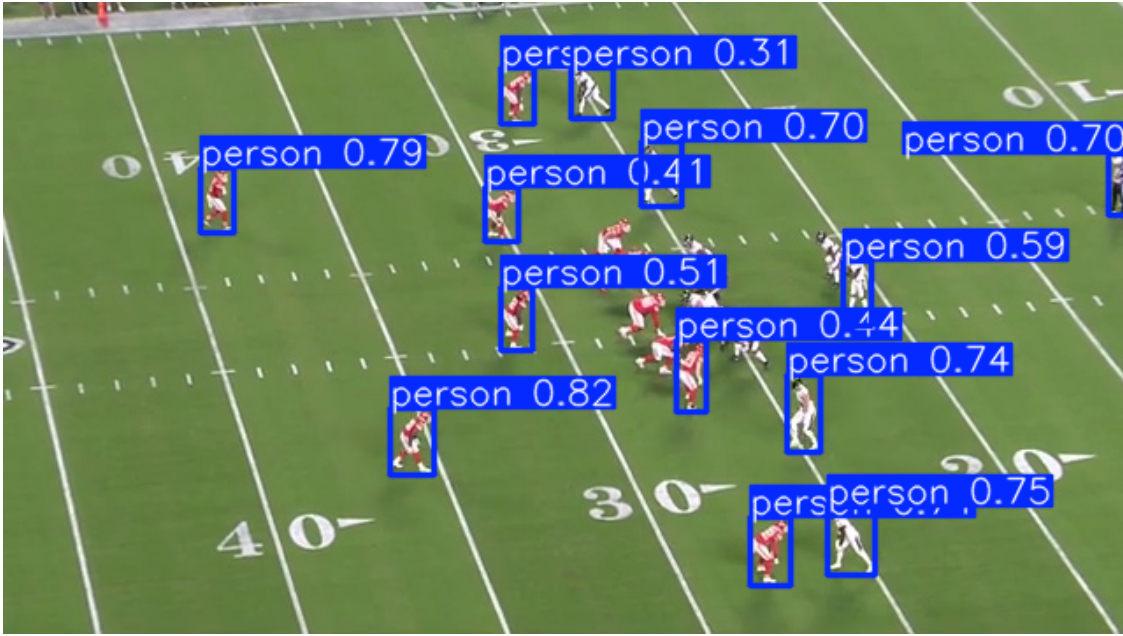
```
[8]: results = model('presnaps-large-model-1/train/images/
      ↪57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg',
      ↪conf=0.25)
      results[0].show()
```

image 1/1 /content/presnaps-large-model-1/train/images/57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg: 640x640 19 persons, 1 tv, 10.0ms
Speed: 2.5ms preprocess, 10.0ms inference, 1.6ms postprocess per image at shape (1, 3, 640, 640)



```
[9]: # Check model on newly acquired presnap screenshots
new_result = model('/content/drive/MyDrive/CU-Boulder-MSCS/Machine_Learning/
↳Deep_Learning/Final_Project/pass_0001_88.jpg', conf=.25)
new_result[0].show()
```

```
image 1/1 /content/drive/MyDrive/CU-Boulder-
MSCS/Machine_Learning/Deep_Learning/Final_Project/pass_0001_88.jpg: 384x640 13
persons, 60.0ms
Speed: 2.4ms preprocess, 60.0ms inference, 2.2ms postprocess per image at shape
(1, 3, 384, 640)
```



```
[16]: # Fine-tune on your football dataset
model.train(
    data='presnaps-large-model-1/data.yaml',
    epochs=50,
    imgsz=640,
    batch=16,
    name='football_detector'
)
```

Ultralytics 8.3.155 Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (NVIDIA A100-SXM4-40GB, 40507MiB)

engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=presnaps-large-model-1/data.yaml, degrees=0.0, deterministic=True, device=None, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=50, erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolov8m.pt, momentum=0.937, mosaic=1.0, multi_scale=False, name=football_detector5, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, retina_masks=False, save=True, save_conf=False, save_crop=False, save_dir=runs/detect/football_detector5, save_frames=False, save_json=False,

```

save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False,
show_boxes=True, show_conf=True, show_labels=True, simplify=True,
single_cls=False, source=None, split=val, stream_buffer=False, task=detect,
time=None, tracker=botssort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.0, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

```

	from	n	params	module
arguments				
0	-1	1	1392	ultralytics.nn.modules.conv.Conv
[3, 48, 3, 2]				
1	-1	1	41664	ultralytics.nn.modules.conv.Conv
[48, 96, 3, 2]				
2	-1	2	111360	ultralytics.nn.modules.block.C2f
[96, 96, 2, True]				
3	-1	1	166272	ultralytics.nn.modules.conv.Conv
[96, 192, 3, 2]				
4	-1	4	813312	ultralytics.nn.modules.block.C2f
[192, 192, 4, True]				
5	-1	1	664320	ultralytics.nn.modules.conv.Conv
[192, 384, 3, 2]				
6	-1	4	3248640	ultralytics.nn.modules.block.C2f
[384, 384, 4, True]				
7	-1	1	1991808	ultralytics.nn.modules.conv.Conv
[384, 576, 3, 2]				
8	-1	2	3985920	ultralytics.nn.modules.block.C2f
[576, 576, 2, True]				
9	-1	1	831168	ultralytics.nn.modules.block.SPPF
[576, 576, 5]				
10	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
12	-1	2	1993728	ultralytics.nn.modules.block.C2f
[960, 384, 2]				
13	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
15	-1	2	517632	ultralytics.nn.modules.block.C2f
[576, 192, 2]				
16	-1	1	332160	ultralytics.nn.modules.conv.Conv
[192, 192, 3, 2]				
17	[-1, 12]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
18	-1	2	1846272	ultralytics.nn.modules.block.C2f
[576, 384, 2]				
19	-1	1	1327872	ultralytics.nn.modules.conv.Conv


```

[384, 384, 3, 2]
 20          [-1, 9]  1          0  ultralytics.nn.modules.conv.Concat
[1]
 21          -1  2   4207104  ultralytics.nn.modules.block.C2f
[960, 576, 2]
 22      [15, 18, 21]  1   3779749  ultralytics.nn.modules.head.Detect
[7, [192, 384, 576]]
Model summary: 169 layers, 25,860,373 parameters, 25,860,357 gradients, 79.1
GFLOPs

Transferred 475/475 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access  (ping: 0.0±0.0 ms, read: 1629.2±506.3
MB/s, size: 50.5 KB)

train: Scanning /content/presnaps-large-
model-1/train/labels.cache... 200 images, 0 backgrounds, 0 corrupt:
100%|          | 200/200 [00:00<?, ?it/s]

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8,
8))

val: Fast image access  (ping: 0.0±0.0 ms, read: 329.7±84.6 MB/s,
size: 43.1 KB)

val: Scanning /content/presnaps-large-model-1/valid/labels.cache...
50 images, 0 backgrounds, 0 corrupt: 100%|          | 50/50 [00:00<?, ?it/s]

Plotting labels to runs/detect/football_detector5/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and
'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
automatically...
optimizer: AdamW(lr=0.000909, momentum=0.9) with parameter groups
77 weight(decay=0.0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to runs/detect/football_detector5
Starting training for 50 epochs...

      Epoch    GPU_mem  box_loss  cls_loss  dfl_loss  Instances    Size
      1/50      7.27G      1.679      1.139      1.226        338      640:
100%|          | 13/13 [00:02<00:00,  5.33it/s]
      Class      Images  Instances  Box(P          R      mAP50
mAP50-95): 100%|          | 2/2 [00:00<00:00,  5.78it/s]

```


0.356	all	50	1157	0.556	0.596	0.637
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	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	2/50	8.4G	1.688	1.097	1.203	328	640:
100%	13/13	[00:01<00:00, 7.48it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95):	100%	2/2	[00:00<00:00, 6.26it/s]				
	all	50	1157	0.502	0.519	0.566	
0.283							

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	3/50	8.43G	1.596	1.098	1.188	296	640:
100%	13/13	[00:01<00:00, 7.60it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95):	100%	2/2	[00:00<00:00, 6.23it/s]				
	all	50	1157	0.528	0.661	0.63	
0.359							

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	4/50	8.43G	1.624	1.101	1.221	233	640:
100%	13/13	[00:01<00:00, 7.72it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95):	100%	2/2	[00:00<00:00, 6.26it/s]				
	all	50	1157	0.578	0.653	0.661	
0.385							

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	5/50	8.43G	1.592	1.053	1.175	263	640:
100%	13/13	[00:01<00:00, 7.92it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95):	100%	2/2	[00:00<00:00, 6.09it/s]				
	all	50	1157	0.564	0.731	0.687	
0.419							

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
6/50	8.5G	1.613	1.028	1.197	398	640:
100%	13/13	[00:01<00:00, 7.97it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.17it/s]				
	all	50	1157	0.554	0.627	0.605
0.339						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
7/50	8.5G	1.632	1.007	1.198	348	640:
100%	13/13	[00:01<00:00, 8.21it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.29it/s]				
	all	50	1157	0.619	0.673	0.676
0.405						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
8/50	8.57G	1.6	1.015	1.181	205	640:
100%	13/13	[00:01<00:00, 7.97it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.37it/s]				
	all	50	1157	0.66	0.673	0.717
0.433						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
9/50	8.63G	1.575	1	1.169	243	640:
100%	13/13	[00:01<00:00, 8.19it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.37it/s]				
	all	50	1157	0.691	0.665	0.745
0.435						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
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```

10/50      8.7G      1.615      0.9973      1.185      336      640:
100%|      | 13/13 [00:01<00:00, 8.16it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.43it/s]
          all          50      1157      0.656      0.714      0.736
0.455

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances      Size
11/50      8.77G          1.62          0.963          1.182          250          640:
100%|      | 13/13 [00:01<00:00, 8.18it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.40it/s]
          all          50      1157          0.66      0.697      0.73
0.448

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances      Size
12/50      8.77G          1.609          0.9431          1.169          243          640:
100%|      | 13/13 [00:01<00:00, 8.00it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.57it/s]
          all          50      1157          0.658      0.665      0.698
0.378

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances      Size
13/50      8.77G          1.612          0.9503          1.177          143          640:
100%|      | 13/13 [00:01<00:00, 8.26it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.49it/s]
          all          50      1157          0.71      0.748      0.788
0.509

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances      Size
14/50      8.77G          1.583          0.9381          1.157          254          640:
100%|      | 13/13 [00:01<00:00, 7.99it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.42it/s]

```

0.429	all	50	1157	0.716	0.718	0.773
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	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	15/50	8.77G	1.545	0.9152	1.159	267	640:
100%	13/13	[00:01<00:00, 6.91it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	2/2	[00:00<00:00, 6.54it/s]					
	all	50	1157	0.726	0.733	0.784	

0.461

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	16/50	8.77G	1.597	0.9408	1.185	314	640:
100%	13/13	[00:01<00:00, 7.95it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	2/2	[00:00<00:00, 6.47it/s]					
	all	50	1157	0.67	0.656	0.721	

0.441

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	17/50	8.77G	1.498	0.9156	1.141	249	640:
100%	13/13	[00:01<00:00, 8.08it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	2/2	[00:00<00:00, 6.36it/s]					
	all	50	1157	0.741	0.729	0.767	

0.464

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	18/50	8.77G	1.473	0.8745	1.134	204	640:
100%	13/13	[00:01<00:00, 8.19it/s]					
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	2/2	[00:00<00:00, 6.28it/s]					
	all	50	1157	0.705	0.749	0.789	

0.455

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
19/50	8.77G	1.477	0.8682	1.131	132	640:
100%	13/13	[00:01<00:00, 8.05it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.06it/s]				
	all	50	1157	0.757	0.73	0.812
0.517						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
20/50	8.77G	1.493	0.8705	1.118	336	640:
100%	13/13	[00:01<00:00, 8.06it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.53it/s]				
	all	50	1157	0.731	0.735	0.801
0.487						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
21/50	8.77G	1.49	0.8858	1.143	224	640:
100%	13/13	[00:01<00:00, 8.17it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.49it/s]				
	all	50	1157	0.72	0.819	0.846
0.501						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
22/50	8.77G	1.501	0.8434	1.127	307	640:
100%	13/13	[00:01<00:00, 8.12it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.36it/s]				
	all	50	1157	0.769	0.734	0.802
0.489						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
-------	---------	----------	----------	----------	-----------	------

```

23/50      8.77G      1.482      0.8555      1.124      327      640:
100%|      | 13/13 [00:01<00:00, 8.01it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.43it/s]
          all          50      1157      0.756      0.773      0.835
0.528

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss  Instances      Size
24/50      8.77G      1.472      0.8423      1.137      207      640:
100%|      | 13/13 [00:01<00:00, 8.11it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.61it/s]
          all          50      1157      0.735      0.835      0.842
0.551

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss  Instances      Size
25/50      8.77G      1.465      0.8376      1.119      190      640:
100%|      | 13/13 [00:01<00:00, 8.29it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.46it/s]
          all          50      1157      0.769      0.787      0.844
0.544

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss  Instances      Size
26/50      8.77G      1.424      0.8128      1.104      154      640:
100%|      | 13/13 [00:01<00:00, 8.03it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.43it/s]
          all          50      1157      0.734      0.755      0.804
0.472

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss  Instances      Size
27/50      8.77G      1.477      0.8251      1.121      166      640:
100%|      | 13/13 [00:01<00:00, 8.18it/s]
          Class      Images  Instances      Box(P          R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.63it/s]

```

	all	50	1157	0.74	0.763	0.812
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0.455

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
28/50	8.77G	1.428	0.7979	1.109	339	640:
100%	13/13	[00:01<00:00, 7.88it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.75it/s]				
	all	50	1157	0.761	0.775	0.836

0.545

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
29/50	8.77G	1.457	0.8218	1.118	262	640:
100%	13/13	[00:01<00:00, 8.16it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.64it/s]				
	all	50	1157	0.796	0.761	0.827

0.519

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
30/50	8.77G	1.432	0.7905	1.098	289	640:
100%	13/13	[00:01<00:00, 8.23it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.39it/s]				
	all	50	1157	0.779	0.785	0.835

0.53

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
31/50	8.77G	1.449	0.788	1.093	298	640:
100%	13/13	[00:01<00:00, 8.15it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.45it/s]				
	all	50	1157	0.747	0.791	0.815

0.485

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
32/50	8.77G	1.42	0.783	1.106	282	640:
100%	13/13	[00:01<00:00, 7.79it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.51it/s]				
	all	50	1157	0.765	0.781	0.809
0.513						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
33/50	8.77G	1.404	0.7677	1.089	283	640:
100%	13/13	[00:01<00:00, 8.26it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.48it/s]				
	all	50	1157	0.743	0.773	0.809
0.502						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
34/50	8.77G	1.362	0.7699	1.081	247	640:
100%	13/13	[00:01<00:00, 8.24it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.67it/s]				
	all	50	1157	0.737	0.806	0.832
0.542						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
35/50	8.77G	1.354	0.7517	1.076	249	640:
100%	13/13	[00:01<00:00, 8.16it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.79it/s]				
	all	50	1157	0.736	0.801	0.831
0.529						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
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36/50      8.77G      1.399      0.7624      1.093      212      640:
100%|      | 13/13 [00:01<00:00, 8.07it/s]
          Class      Images Instances      Box(P      R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.70it/s]
          all          50      1157      0.788      0.793      0.862
0.55

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss Instances      Size
37/50      8.77G      1.389      0.7629      1.087      185      640:
100%|      | 13/13 [00:01<00:00, 8.23it/s]
          Class      Images Instances      Box(P      R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.14it/s]
          all          50      1157      0.788      0.795      0.848
0.535

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss Instances      Size
38/50      8.77G      1.353      0.7451      1.07      220      640:
100%|      | 13/13 [00:01<00:00, 8.20it/s]
          Class      Images Instances      Box(P      R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.71it/s]
          all          50      1157      0.765      0.8      0.833
0.532

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss Instances      Size
39/50      8.77G      1.327      0.7385      1.067      246      640:
100%|      | 13/13 [00:01<00:00, 8.19it/s]
          Class      Images Instances      Box(P      R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.71it/s]
          all          50      1157      0.716      0.828      0.846
0.543

```

```

Epoch      GPU_mem      box_loss      cls_loss      dfl_loss Instances      Size
40/50      8.77G      1.339      0.7376      1.066      301      640:
100%|      | 13/13 [00:01<00:00, 8.04it/s]
          Class      Images Instances      Box(P      R      mAP50
mAP50-95): 100%|      | 2/2 [00:00<00:00, 6.55it/s]

```

	all	50	1157	0.8	0.76	0.843
--	-----	----	------	-----	------	-------

0.53

Closing dataloader mosaic

`albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))`

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
41/50	8.77G	1.333	0.6946	1.107	178	640:
100%	13/13	[00:02<00:00, 5.82it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.53it/s]				
	all	50	1157	0.794	0.772	0.851

0.545

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
42/50	8.77G	1.347	0.6921	1.102	187	640:
100%	13/13	[00:01<00:00, 8.37it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.72it/s]				
	all	50	1157	0.792	0.773	0.855

0.557

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
43/50	8.77G	1.289	0.6735	1.087	185	640:
100%	13/13	[00:01<00:00, 8.31it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.65it/s]				
	all	50	1157	0.777	0.822	0.855

0.563

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
44/50	8.77G	1.277	0.6577	1.075	180	640:
100%	13/13	[00:01<00:00, 8.14it/s]				

	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00,	6.78it/s]			
	all	50	1157	0.782	0.817	0.857

0.561

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
45/50	8.77G	1.281	0.6527	1.079	184	640:
100%	13/13	[00:01<00:00,	8.36it/s]			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00,	6.54it/s]			
	all	50	1157	0.813	0.789	0.855

0.553

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
46/50	8.77G	1.281	0.6556	1.07	187	640:
100%	13/13	[00:01<00:00,	8.13it/s]			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00,	6.10it/s]			
	all	50	1157	0.816	0.781	0.855

0.553

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
47/50	8.77G	1.233	0.6366	1.065	179	640:
100%	13/13	[00:01<00:00,	8.36it/s]			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00,	6.61it/s]			
	all	50	1157	0.803	0.804	0.855

0.548

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
48/50	8.77G	1.263	0.6471	1.063	185	640:
100%	13/13	[00:01<00:00,	8.17it/s]			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00,	6.66it/s]			

	all	50	1157	0.815	0.803	0.863
--	-----	----	------	-------	-------	-------

0.556

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
49/50	8.77G	1.261	0.6428	1.069	184	640:
100%	13/13	[00:01<00:00, 8.48it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.76it/s]				
	all	50	1157	0.812	0.803	0.863

0.554

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
50/50	8.77G	1.254	0.6369	1.052	186	640:
100%	13/13	[00:01<00:00, 8.26it/s]				
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 6.73it/s]				
	all	50	1157	0.811	0.808	0.864

0.554

50 epochs completed in 0.038 hours.

Optimizer stripped from runs/detect/football_detector5/weights/last.pt, 52.0MB

Optimizer stripped from runs/detect/football_detector5/weights/best.pt, 52.0MB

Validating runs/detect/football_detector5/weights/best.pt...

Ultralytics 8.3.155 Python-3.11.13 torch-2.6.0+cu124 CUDA:0 (NVIDIA

A100-SXM4-40GB, 40507MiB)

Model summary (fused): 92 layers, 25,843,813 parameters, 0 gradients, 78.7

GFLOPs

	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	2/2	[00:00<00:00, 5.36it/s]				
	all	50	1157	0.777	0.823	0.855
0.563	defense	50	550	0.908	0.942	0.972
0.671	oline	50	250	0.848	0.924	0.93
0.572	qb	50	50	0.731	0.816	0.871
0.483	ref	31	57	0.808	0.887	0.878

```

0.562
    running_back      44      49      0.747      0.837      0.886
0.581
    tight_end        38      39      0.565      0.513      0.555
0.398
    wide_receiver    50     162      0.83      0.844      0.895
0.671
Speed: 0.1ms preprocess, 1.9ms inference, 0.0ms loss, 1.7ms postprocess per
image
Results saved to runs/detect/football_detector5

```

[16]: ultralytics.utils.metrics.DetMetrics object with attributes:

```

ap_class_index: array([0, 1, 2, 3, 4, 5, 6])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at
0x7feda09c1450>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)',
'Recall-Confidence(B)']
curves_results: [[array([
0, 0.001001, 0.002002, 0.003003,
0.004004, 0.005005, 0.006006, 0.007007, 0.008008, 0.009009,
0.01001, 0.011011, 0.012012, 0.013013, 0.014014, 0.015015,
0.016016, 0.017017, 0.018018, 0.019019, 0.02002, 0.021021,
0.022022, 0.023023,
0.024024, 0.025025, 0.026026, 0.027027, 0.028028,
0.029029, 0.03003, 0.031031, 0.032032, 0.033033, 0.034034,
0.035035, 0.036036, 0.037037, 0.038038, 0.039039, 0.04004,
0.041041, 0.042042, 0.043043, 0.044044, 0.045045, 0.046046,
0.047047,
0.048048, 0.049049, 0.05005, 0.051051, 0.052052,
0.053053, 0.054054, 0.055055, 0.056056, 0.057057, 0.058058,
0.059059, 0.06006, 0.061061, 0.062062, 0.063063, 0.064064,
0.065065, 0.066066, 0.067067, 0.068068, 0.069069, 0.07007,
0.071071,
0.072072, 0.073073, 0.074074, 0.075075, 0.076076,
0.077077, 0.078078, 0.079079, 0.08008, 0.081081, 0.082082,
0.083083, 0.084084, 0.085085, 0.086086, 0.087087, 0.088088,
0.089089, 0.09009, 0.091091, 0.092092, 0.093093, 0.094094,
0.095095,
0.096096, 0.097097, 0.098098, 0.099099, 0.1001,
0.1011, 0.1021, 0.1031, 0.1041, 0.10511, 0.10611,
0.10711, 0.10811, 0.10911, 0.11011, 0.11111, 0.11211,
0.11311, 0.11411, 0.11512, 0.11612, 0.11712, 0.11812,
0.11912,
0.12012, 0.12112, 0.12212, 0.12312, 0.12412,
0.12513, 0.12613, 0.12713, 0.12813, 0.12913, 0.13013,
0.13113, 0.13213, 0.13313, 0.13413, 0.13514, 0.13614,

```

0.13714,	0.13814,	0.13914,	0.14014,	0.14114,	0.14214,
0.14314,					
	0.14414,	0.14515,	0.14615,	0.14715,	0.14815,
0.14915,	0.15015,	0.15115,	0.15215,	0.15315,	0.15415,
0.15516,	0.15616,	0.15716,	0.15816,	0.15916,	0.16016,
0.16116,	0.16216,	0.16316,	0.16416,	0.16517,	0.16617,
0.16717,					
	0.16817,	0.16917,	0.17017,	0.17117,	0.17217,
0.17317,	0.17417,	0.17518,	0.17618,	0.17718,	0.17818,
0.17918,	0.18018,	0.18118,	0.18218,	0.18318,	0.18418,
0.18519,	0.18619,	0.18719,	0.18819,	0.18919,	0.19019,
0.19119,					
	0.19219,	0.19319,	0.19419,	0.1952,	0.1962,
0.1972,	0.1982,	0.1992,	0.2002,	0.2012,	0.2022,
0.2032,	0.2042,	0.20521,	0.20621,	0.20721,	0.20821,
0.20921,	0.21021,	0.21121,	0.21221,	0.21321,	0.21421,
0.21522,					
	0.21622,	0.21722,	0.21822,	0.21922,	0.22022,
0.22122,	0.22222,	0.22322,	0.22422,	0.22523,	0.22623,
0.22723,	0.22823,	0.22923,	0.23023,	0.23123,	0.23223,
0.23323,	0.23423,	0.23524,	0.23624,	0.23724,	0.23824,
0.23924,					
	0.24024,	0.24124,	0.24224,	0.24324,	0.24424,
0.24525,	0.24625,	0.24725,	0.24825,	0.24925,	0.25025,
0.25125,	0.25225,	0.25325,	0.25425,	0.25526,	0.25626,
0.25726,	0.25826,	0.25926,	0.26026,	0.26126,	0.26226,
0.26326,					
	0.26426,	0.26527,	0.26627,	0.26727,	0.26827,
0.26927,	0.27027,	0.27127,	0.27227,	0.27327,	0.27427,
0.27528,	0.27628,	0.27728,	0.27828,	0.27928,	0.28028,
0.28128,	0.28228,	0.28328,	0.28428,	0.28529,	0.28629,
0.28729,					
	0.28829,	0.28929,	0.29029,	0.29129,	0.29229,
0.29329,	0.29429,	0.2953,	0.2963,	0.2973,	0.2983,
0.2993,	0.3003,	0.3013,	0.3023,	0.3033,	0.3043,
0.30531,	0.30631,	0.30731,	0.30831,	0.30931,	0.31031,
0.31131,					
	0.31231,	0.31331,	0.31431,	0.31532,	0.31632,
0.31732,	0.31832,	0.31932,	0.32032,	0.32132,	0.32232,
0.32332,	0.32432,	0.32533,	0.32633,	0.32733,	0.32833,
0.32933,	0.33033,	0.33133,	0.33233,	0.33333,	0.33433,
0.33534,					
	0.33634,	0.33734,	0.33834,	0.33934,	0.34034,
0.34134,	0.34234,	0.34334,	0.34434,	0.34535,	0.34635,
0.34735,	0.34835,	0.34935,	0.35035,	0.35135,	0.35235,
0.35335,	0.35435,	0.35536,	0.35636,	0.35736,	0.35836,
0.35936,					

	0.36036,	0.36136,	0.36236,	0.36336,	0.36436,
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0.55756,	0.55856,	0.55956,	0.56056,	0.56156,	0.56256,
0.56356,	0.56456,	0.56557,	0.56657,	0.56757,	0.56857,
0.56957,	0.57057,	0.57157,	0.57257,	0.57357,	0.57457,
0.57558,	0.57658,	0.57758,	0.57858,	0.57958,	0.58058,
0.58158,	0.58258,	0.58358,	0.58458,	0.58559,	0.58659,
0.58759,	0.58859,	0.58959,	0.59059,	0.59159,	0.59259,
0.59359,	0.59459,	0.5956,	0.5966,	0.5976,	0.5986,
0.5996,	0.6006,	0.6016,	0.6026,	0.6036,	0.6046,
0.60561,	0.60661,	0.60761,	0.60861,	0.60961,	0.61061,
0.61161,	0.61261,	0.61361,	0.61461,	0.61562,	0.61662,
0.61762,	0.61862,	0.61962,	0.62062,	0.62162,	0.62262,
0.62362,	0.62462,	0.62563,	0.62663,	0.62763,	0.62863,
0.62963,	0.63063,	0.63163,	0.63263,	0.63363,	0.63463,
0.63564,	0.63664,	0.63764,	0.63864,	0.63964,	0.64064,
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0.65966,	0.66066,	0.66166,	0.66266,	0.66366,	0.66466,

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	0.67267,	0.67367,	0.67467,	0.67568,	0.67668,
0.67768,	0.67868,	0.67968,	0.68068,	0.68168,	0.68268,
0.68368,	0.68468,	0.68569,	0.68669,	0.68769,	0.68869,
0.68969,	0.69069,	0.69169,	0.69269,	0.69369,	0.69469,
0.6957,					
	0.6967,	0.6977,	0.6987,	0.6997,	0.7007,
0.7017,	0.7027,	0.7037,	0.7047,	0.70571,	0.70671,
0.70771,	0.70871,	0.70971,	0.71071,	0.71171,	0.71271,
0.71371,	0.71471,	0.71572,	0.71672,	0.71772,	0.71872,
0.71972,					
	0.72072,	0.72172,	0.72272,	0.72372,	0.72472,
0.72573,	0.72673,	0.72773,	0.72873,	0.72973,	0.73073,
0.73173,	0.73273,	0.73373,	0.73473,	0.73574,	0.73674,
0.73774,	0.73874,	0.73974,	0.74074,	0.74174,	0.74274,
0.74374,					
	0.74474,	0.74575,	0.74675,	0.74775,	0.74875,
0.74975,	0.75075,	0.75175,	0.75275,	0.75375,	0.75475,
0.75576,	0.75676,	0.75776,	0.75876,	0.75976,	0.76076,
0.76176,	0.76276,	0.76376,	0.76476,	0.76577,	0.76677,
0.76777,					
	0.76877,	0.76977,	0.77077,	0.77177,	0.77277,
0.77377,	0.77477,	0.77578,	0.77678,	0.77778,	0.77878,
0.77978,	0.78078,	0.78178,	0.78278,	0.78378,	0.78478,
0.78579,	0.78679,	0.78779,	0.78879,	0.78979,	0.79079,
0.79179,					
	0.79279,	0.79379,	0.79479,	0.7958,	0.7968,
0.7978,	0.7988,	0.7998,	0.8008,	0.8018,	0.8028,
0.8038,	0.8048,	0.80581,	0.80681,	0.80781,	0.80881,
0.80981,	0.81081,	0.81181,	0.81281,	0.81381,	0.81481,
0.81582,					
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0.82783,	0.82883,	0.82983,	0.83083,	0.83183,	0.83283,
0.83383,	0.83483,	0.83584,	0.83684,	0.83784,	0.83884,
0.83984,					
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0.84585,	0.84685,	0.84785,	0.84885,	0.84985,	0.85085,
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0.85786,	0.85886,	0.85986,	0.86086,	0.86186,	0.86286,
0.86386,					
	0.86486,	0.86587,	0.86687,	0.86787,	0.86887,
0.86987,	0.87087,	0.87187,	0.87287,	0.87387,	0.87487,
0.87588,	0.87688,	0.87788,	0.87888,	0.87988,	0.88088,
0.88188,	0.88288,	0.88388,	0.88488,	0.88589,	0.88689,
0.88789,					

	0.88889,	0.88989,	0.89089,	0.89189,	0.89289,
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0.8999,	0.9009,	0.9019,	0.9029,	0.9039,	0.9049,
0.90591,	0.90691,	0.90791,	0.90891,	0.90991,	0.91091,
0.91191,					
	0.91291,	0.91391,	0.91491,	0.91592,	0.91692,
0.91792,	0.91892,	0.91992,	0.92092,	0.92192,	0.92292,
0.92392,	0.92492,	0.92593,	0.92693,	0.92793,	0.92893,
0.92993,	0.93093,	0.93193,	0.93293,	0.93393,	0.93493,
0.93594,					
	0.93694,	0.93794,	0.93894,	0.93994,	0.94094,
0.94194,	0.94294,	0.94394,	0.94494,	0.94595,	0.94695,
0.94795,	0.94895,	0.94995,	0.95095,	0.95195,	0.95295,
0.95395,	0.95495,	0.95596,	0.95696,	0.95796,	0.95896,
0.95996,					
	0.96096,	0.96196,	0.96296,	0.96396,	0.96496,
0.96597,	0.96697,	0.96797,	0.96897,	0.96997,	0.97097,
0.97197,	0.97297,	0.97397,	0.97497,	0.97598,	0.97698,
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0.98398,					
	0.98498,	0.98599,	0.98699,	0.98799,	0.98899,
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0.021021,	0.022022,	0.023023,			
	0.024024,	0.025025,	0.026026,	0.027027,	0.028028,
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0.035035,	0.036036,	0.037037,	0.038038,	0.039039,	0.04004,
0.041041,	0.042042,	0.043043,	0.044044,	0.045045,	0.046046,
0.047047,					
	0.048048,	0.049049,	0.05005,	0.051051,	0.052052,
0.053053,	0.054054,	0.055055,	0.056056,	0.057057,	0.058058,
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0.071071,	0.072072,	0.073073,	0.074074,	0.075075,	0.076076,
0.077077,	0.078078,	0.079079,	0.08008,	0.081081,	0.082082,
0.083083,	0.084084,	0.085085,	0.086086,	0.087087,	0.088088,
0.089089,	0.09009,	0.091091,	0.092092,	0.093093,	0.094094,
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0.13113,	0.13213,	0.13313,	0.13413,	0.13514,	0.13614,
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0.15516,	0.15616,	0.15716,	0.15816,	0.15916,	0.16016,
0.16116,	0.16216,	0.16316,	0.16416,	0.16517,	0.16617,
0.16717,	0.16817,	0.16917,	0.17017,	0.17117,	0.17217,
0.17317,	0.17417,	0.17518,	0.17618,	0.17718,	0.17818,
0.17918,	0.18018,	0.18118,	0.18218,	0.18318,	0.18418,
0.18519,	0.18619,	0.18719,	0.18819,	0.18919,	0.19019,
0.19119,	0.19219,	0.19319,	0.19419,	0.1952,	0.1962,
0.1972,	0.1982,	0.1992,	0.2002,	0.2012,	0.2022,
0.2032,	0.2042,	0.20521,	0.20621,	0.20721,	0.20821,
0.20921,	0.21021,	0.21121,	0.21221,	0.21321,	0.21421,
0.21522,	0.21622,	0.21722,	0.21822,	0.21922,	0.22022,
0.22122,	0.22222,	0.22322,	0.22422,	0.22523,	0.22623,
0.22723,	0.22823,	0.22923,	0.23023,	0.23123,	0.23223,
0.23323,	0.23423,	0.23524,	0.23624,	0.23724,	0.23824,
0.23924,	0.24024,	0.24124,	0.24224,	0.24324,	0.24424,
0.24525,	0.24625,	0.24725,	0.24825,	0.24925,	0.25025,
0.25125,	0.25225,	0.25325,	0.25425,	0.25526,	0.25626,
0.25726,	0.25826,	0.25926,	0.26026,	0.26126,	0.26226,
0.26326,	0.26426,	0.26527,	0.26627,	0.26727,	0.26827,
0.26927,	0.27027,	0.27127,	0.27227,	0.27327,	0.27427,
0.27528,	0.27628,	0.27728,	0.27828,	0.27928,	0.28028,
0.28128,	0.28228,	0.28328,	0.28428,	0.28529,	0.28629,
0.28729,					

	0.28829,	0.28929,	0.29029,	0.29129,	0.29229,
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0.31131,					
	0.31231,	0.31331,	0.31431,	0.31532,	0.31632,
0.31732,	0.31832,	0.31932,	0.32032,	0.32132,	0.32232,
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0.32933,	0.33033,	0.33133,	0.33233,	0.33333,	0.33433,
0.33534,					
	0.33634,	0.33734,	0.33834,	0.33934,	0.34034,
0.34134,	0.34234,	0.34334,	0.34434,	0.34535,	0.34635,
0.34735,	0.34835,	0.34935,	0.35035,	0.35135,	0.35235,
0.35335,	0.35435,	0.35536,	0.35636,	0.35736,	0.35836,
0.35936,					
	0.36036,	0.36136,	0.36236,	0.36336,	0.36436,
0.36537,	0.36637,	0.36737,	0.36837,	0.36937,	0.37037,
0.37137,	0.37237,	0.37337,	0.37437,	0.37538,	0.37638,
0.37738,	0.37838,	0.37938,	0.38038,	0.38138,	0.38238,
0.38338,					
	0.38438,	0.38539,	0.38639,	0.38739,	0.38839,
0.38939,	0.39039,	0.39139,	0.39239,	0.39339,	0.39439,
0.3954,	0.3964,	0.3974,	0.3984,	0.3994,	0.4004,
0.4014,	0.4024,	0.4034,	0.4044,	0.40541,	0.40641,
0.40741,					
	0.40841,	0.40941,	0.41041,	0.41141,	0.41241,
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0.41942,	0.42042,	0.42142,	0.42242,	0.42342,	0.42442,
0.42543,	0.42643,	0.42743,	0.42843,	0.42943,	0.43043,
0.43143,					
	0.43243,	0.43343,	0.43443,	0.43544,	0.43644,
0.43744,	0.43844,	0.43944,	0.44044,	0.44144,	0.44244,
0.44344,	0.44444,	0.44545,	0.44645,	0.44745,	0.44845,
0.44945,	0.45045,	0.45145,	0.45245,	0.45345,	0.45445,
0.45546,					
	0.45646,	0.45746,	0.45846,	0.45946,	0.46046,
0.46146,	0.46246,	0.46346,	0.46446,	0.46547,	0.46647,
0.46747,	0.46847,	0.46947,	0.47047,	0.47147,	0.47247,
0.47347,	0.47447,	0.47548,	0.47648,	0.47748,	0.47848,
0.47948,					
	0.48048,	0.48148,	0.48248,	0.48348,	0.48448,
0.48549,	0.48649,	0.48749,	0.48849,	0.48949,	0.49049,
0.49149,	0.49249,	0.49349,	0.49449,	0.4955,	0.4965,
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0.5035,					
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0.50951,	0.51051,	0.51151,	0.51251,	0.51351,	0.51451,

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0.52152,	0.52252,	0.52352,	0.52452,	0.52553,	0.52653,
0.52753,					
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0.53353,	0.53453,	0.53554,	0.53654,	0.53754,	0.53854,
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0.54555,	0.54655,	0.54755,	0.54855,	0.54955,	0.55055,
0.55155,					
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0.55756,	0.55856,	0.55956,	0.56056,	0.56156,	0.56256,
0.56356,	0.56456,	0.56557,	0.56657,	0.56757,	0.56857,
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0.57558,					
	0.57658,	0.57758,	0.57858,	0.57958,	0.58058,
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0.59359,	0.59459,	0.5956,	0.5966,	0.5976,	0.5986,
0.5996,					
	0.6006,	0.6016,	0.6026,	0.6036,	0.6046,
0.60561,	0.60661,	0.60761,	0.60861,	0.60961,	0.61061,
0.61161,	0.61261,	0.61361,	0.61461,	0.61562,	0.61662,
0.61762,	0.61862,	0.61962,	0.62062,	0.62162,	0.62262,
0.62362,					
	0.62462,	0.62563,	0.62663,	0.62763,	0.62863,
0.62963,	0.63063,	0.63163,	0.63263,	0.63363,	0.63463,
0.63564,	0.63664,	0.63764,	0.63864,	0.63964,	0.64064,
0.64164,	0.64264,	0.64364,	0.64464,	0.64565,	0.64665,
0.64765,					
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0.65966,	0.66066,	0.66166,	0.66266,	0.66366,	0.66466,
0.66567,	0.66667,	0.66767,	0.66867,	0.66967,	0.67067,
0.67167,					
	0.67267,	0.67367,	0.67467,	0.67568,	0.67668,
0.67768,	0.67868,	0.67968,	0.68068,	0.68168,	0.68268,
0.68368,	0.68468,	0.68569,	0.68669,	0.68769,	0.68869,
0.68969,	0.69069,	0.69169,	0.69269,	0.69369,	0.69469,
0.6957,					
	0.6967,	0.6977,	0.6987,	0.6997,	0.7007,
0.7017,	0.7027,	0.7037,	0.7047,	0.70571,	0.70671,
0.70771,	0.70871,	0.70971,	0.71071,	0.71171,	0.71271,
0.71371,	0.71471,	0.71572,	0.71672,	0.71772,	0.71872,
0.71972,					
	0.72072,	0.72172,	0.72272,	0.72372,	0.72472,
0.72573,	0.72673,	0.72773,	0.72873,	0.72973,	0.73073,
0.73173,	0.73273,	0.73373,	0.73473,	0.73574,	0.73674,
0.73774,	0.73874,	0.73974,	0.74074,	0.74174,	0.74274,

0.74374,					
	0.74474,	0.74575,	0.74675,	0.74775,	0.74875,
0.74975,	0.75075,	0.75175,	0.75275,	0.75375,	0.75475,
0.75576,	0.75676,	0.75776,	0.75876,	0.75976,	0.76076,
0.76176,	0.76276,	0.76376,	0.76476,	0.76577,	0.76677,
0.76777,					
	0.76877,	0.76977,	0.77077,	0.77177,	0.77277,
0.77377,	0.77477,	0.77578,	0.77678,	0.77778,	0.77878,
0.77978,	0.78078,	0.78178,	0.78278,	0.78378,	0.78478,
0.78579,	0.78679,	0.78779,	0.78879,	0.78979,	0.79079,
0.79179,					
	0.79279,	0.79379,	0.79479,	0.7958,	0.7968,
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0.80981,	0.81081,	0.81181,	0.81281,	0.81381,	0.81481,
0.81582,					
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0.83383,	0.83483,	0.83584,	0.83684,	0.83784,	0.83884,
0.83984,					
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0.86386,					
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0.88789,					
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    0.072072,    0.073073,    0.074074,    0.075075,    0.076076,
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0.21522,					
	0.21622,	0.21722,	0.21822,	0.21922,	0.22022,
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0.23924,					
	0.24024,	0.24124,	0.24224,	0.24324,	0.24424,
0.24525,	0.24625,	0.24725,	0.24825,	0.24925,	0.25025,
0.25125,	0.25225,	0.25325,	0.25425,	0.25526,	0.25626,
0.25726,	0.25826,	0.25926,	0.26026,	0.26126,	0.26226,
0.26326,					
	0.26426,	0.26527,	0.26627,	0.26727,	0.26827,
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0.35936,					
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0.80981,	0.81081,	0.81181,	0.81281,	0.81381,	0.81481,
0.81582,					

	0.81682,	0.81782,	0.81882,	0.81982,	0.82082,
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0.83383,	0.83483,	0.83584,	0.83684,	0.83784,	0.83884,
0.83984,					
	0.84084,	0.84184,	0.84284,	0.84384,	0.84484,
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0.85786,	0.85886,	0.85986,	0.86086,	0.86186,	0.86286,
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0.86987,	0.87087,	0.87187,	0.87287,	0.87387,	0.87487,
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0.98398,					
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0]]), 'Confidence', 'Recall']]
fitness: np.float64(0.5919712673945288)
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)',
'metrics/mAP50-95(B)']
maps: array([ 0.67149, 0.57237, 0.4834, 0.56192, 0.58141,
0.39793, 0.67053])
names: {0: 'defense', 1: 'oline', 2: 'qb', 3: 'ref', 4: 'running_back', 5:
'tight_end', 6: 'wide_receiver'}
nt_per_class: array([550, 250, 50, 57, 49, 39, 162])
nt_per_image: array([50, 50, 50, 31, 44, 38, 50])
results_dict: {'metrics/precision(B)': np.float64(0.7767494032071148),
'metrics/recall(B)': np.float64(0.8231150535356162), 'metrics/mAP50(B)':
np.float64(0.8552213946750722), 'metrics/mAP50-95(B)':
np.float64(0.5627212532522462), 'fitness': np.float64(0.5919712673945288)}
speed: {'preprocess': 0.13473277999764832, 'inference': 1.8757241799994517,
'loss': 0.00032343999919248745, 'postprocess': 1.7486471600022924}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img':
[]}]
task: 'detect'

```

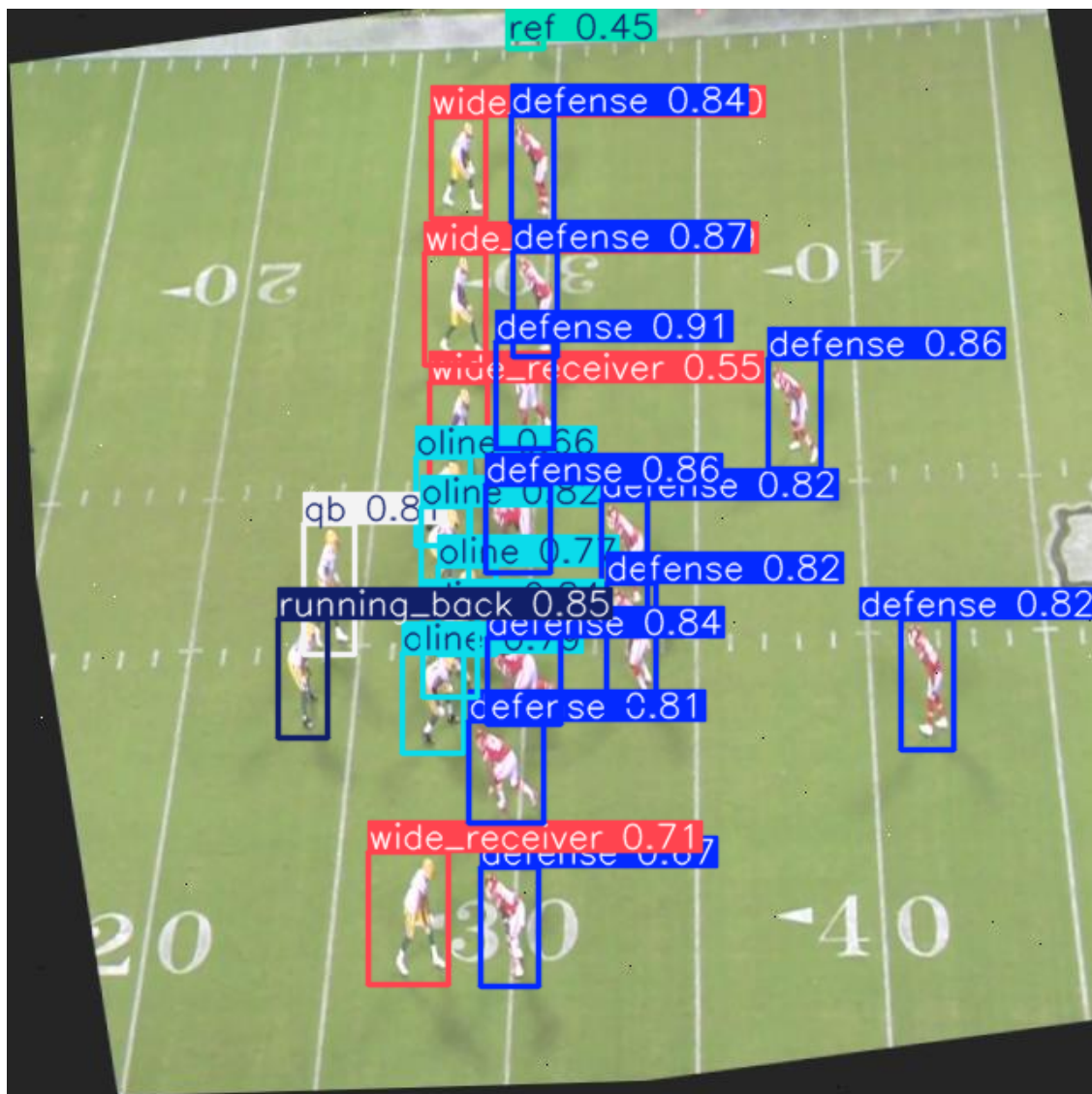
```
[17]: ft_model = YOLO('runs/detect/football_detector5/weights/best.pt')
```

```
[18]: results = ft_model('presnaps-large-model-1/train/images/
↪57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb6fca8.jpg',
↪conf=0.25)
results[0].show()
```

```

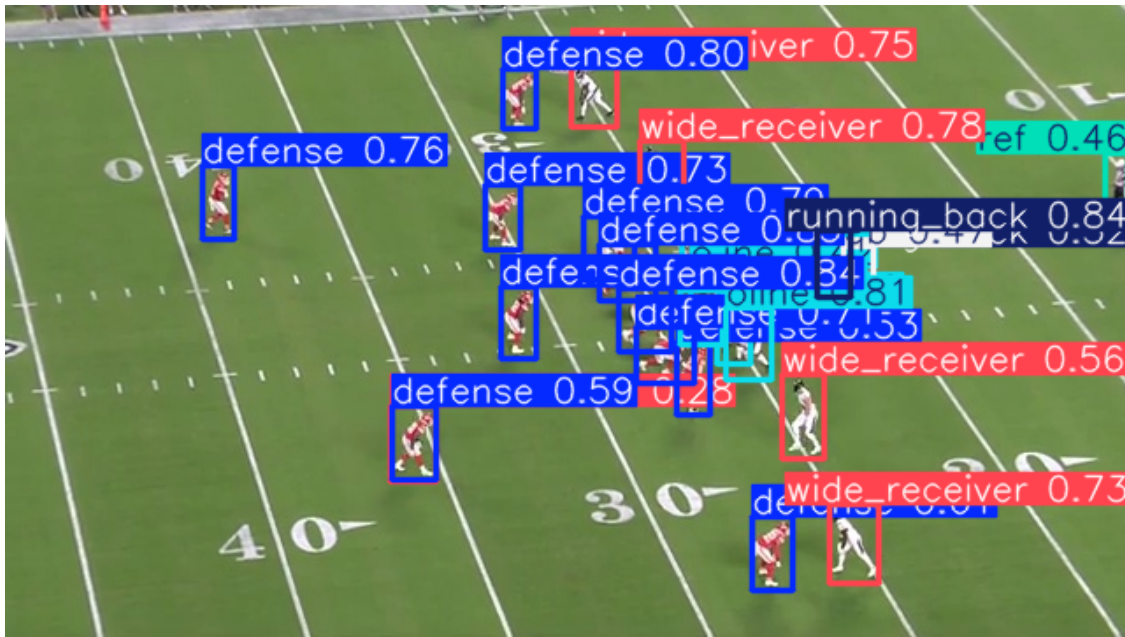
image 1/1 /content/presnaps-large-model-
1/train/images/57558_002886_Sideline_frame204_jpg.rf.386300f508e1beea9a6034e14cb
6fca8.jpg: 640x640 11 defenses, 5 olines, 1 qb, 1 ref, 1 running_back, 4
wide_receivers, 10.2ms
Speed: 2.0ms preprocess, 10.2ms inference, 2.0ms postprocess per image at shape
(1, 3, 640, 640)

```



```
[19]: # Check model on newly acquired presnap screenshots
new_result = ft_model('/content/drive/MyDrive/CU-Boulder-MSCS/Machine_Learning/
↳Deep_Learning/Final_Project/pass_0001_88.jpg', conf=.25)
new_result[0].show()
```

```
image 1/1 /content/drive/MyDrive/CU-Boulder-
MSCS/Machine_Learning/Deep_Learning/Final_Project/pass_0001_88.jpg: 384x640 11
defenses, 5 olines, 1 qb, 1 ref, 2 running_backs, 5 wide_receivers, 14.1ms
Speed: 2.4ms preprocess, 14.1ms inference, 2.3ms postprocess per image at shape
(1, 3, 384, 640)
```



```
[20]: from huggingface_hub import HfApi, create_repo
```

```
# Initialize API
api = HfApi()
```

```
[21]: # Create repository (if it doesn't exist)
repo_id = "lmarte17/gridiron-ai-football-detector"
create_repo(repo_id=repo_id, exist_ok=True)
```

```
[21]: RepoUrl('https://huggingface.co/lmarte17/gridiron-ai-football-detector',
endpoint='https://huggingface.co', repo_type='model',
repo_id='lmarte17/gridiron-ai-football-detector')
```

```
[23]: # Push your trained YOLOv8 model
api.upload_file(
    path_or_fileobj="runs/detect/football_detector5/weights/best.pt", # Your_
    ↪model path
    path_in_repo="best.pt", # Name in the repo
    repo_id=repo_id,
    commit_message="Add trained YOLOv8 football detection model"
)

# Also upload other important files
api.upload_file(
    path_or_fileobj="presnaps-large-model-1/data.yaml",
    path_in_repo="data.yaml",
```

```
repo_id=repo_id,  
commit_message="Add dataset configuration"  
)
```

No files have been modified since last commit. Skipping to prevent empty commit.
WARNING:huggingface_hub.hf_api:No files have been modified since last commit.
Skipping to prevent empty commit.

```
[23]: CommitInfo(commit_url='https://huggingface.co/lmarte17/gridiron-ai-football-  
detector/commit/342e04023ce80b3a7764559d0c9d4926be39e015', commit_message='Add  
dataset configuration', commit_description='',  
oid='342e04023ce80b3a7764559d0c9d4926be39e015', pr_url=None,  
repo_url=RepoUrl('https://huggingface.co/lmarte17/gridiron-ai-football-  
detector', endpoint='https://huggingface.co', repo_type='model',  
repo_id='lmarte17/gridiron-ai-football-detector'), pr_revision=None,  
pr_num=None)
```